

Lecture 8:

Evaluation of a binary classification model

Florent Pollet

BINF4002
MLHC

February 12, 2026



COLUMBIA

COLUMBIA UNIVERSITY
DEPARTMENT OF
BIOMEDICAL INFORMATICS

Acknowledgments

Some slides / figures come from Prof. Matthew McDermott's previous presentations. Thank you!

Outline

1 Hard Classification

- Definition
- Confusion Matrix
- Utility

2 Soft Classification

- Definition
- Score vs Probabilities
- Bayes Optimal Classifier
- Decision Curve Analysis
- Discrimination vs Calibration
- AUROC and its limitations

Part 1

Hard Classification

- 1 Hard Classification
 - Definition
 - Confusion Matrix
 - Utility
- 2 Soft Classification
 - Definition
 - Score vs Probabilities
 - Bayes Optimal Classifier
 - Decision Curve Analysis
 - Discrimination vs Calibration
 - AUROC and its limitations

Deterministic Binary Classification & The Decision Stump

1. The Setup: Hard Classification

- **Goal:** Map input $x \in \mathcal{X}$ to a fixed discrete label $y \in \{0, 1\}$.
- **Model:** A fixed function $h : \mathcal{X} \rightarrow \{0, 1\}$.
 - 0: Negative Class (e.g., Healthy)
 - 1: Positive Class (e.g., Diseased)

Deterministic Binary Classification & The Decision Stump

1. The Setup: Hard Classification

- **Goal:** Map input $x \in \mathcal{X}$ to a fixed discrete label $y \in \{0, 1\}$.
- **Model:** A fixed function $h : \mathcal{X} \rightarrow \{0, 1\}$.
 - 0: Negative Class (e.g., Healthy)
 - 1: Positive Class (e.g., Diseased)

2. Example: The Decision Stump

- A **one-level decision tree**.
- Classification based on a single feature x_j and threshold τ .

Mathematical Form:

$$h(x) = \begin{cases} 1 & \text{if } x_j > \tau \\ 0 & \text{otherwise} \end{cases}$$

Deterministic Binary Classification & The Decision Stump

1. The Setup: Hard Classification

- **Goal:** Map input $x \in \mathcal{X}$ to a fixed discrete label $y \in \{0, 1\}$.
- **Model:** A fixed function $h : \mathcal{X} \rightarrow \{0, 1\}$.
 - 0: Negative Class (e.g., Healthy)
 - 1: Positive Class (e.g., Diseased)

2. Example: The Decision Stump

- A **one-level decision tree**.
- Classification based on a single feature x_j and threshold τ .

Mathematical Form:

$$h(x) = \begin{cases} 1 & \text{if } x_j > \tau \\ 0 & \text{otherwise} \end{cases}$$

Clinical Context

Sepsis Triage Rule:

- **Feature (x_j):** Temp.
- **Threshold (τ):** 38°C
- **Rule:** Temp > 38 $\rightarrow$ Risk

The Confusion Matrix - the traditional way

	Predicted: True (1)	Predicted: False (0)
Actual: True (1)		

The Confusion Matrix - the traditional way

	Predicted: True (1)	Predicted: False (0)
Actual: True (1)	True Positive (TP)	

The Confusion Matrix - the traditional way

	Predicted: True (1)	Predicted: False (0)
Actual: True (1)	True Positive (TP) <i>(Hit)</i>	False Negative (FN) <i>(Type II Error / Miss)</i>
Actual: False (0)		

The Confusion Matrix - the traditional way

	Predicted: True (1)	Predicted: False (0)
Actual: True (1)	True Positive (TP) <i>(Hit)</i>	False Negative (FN) <i>(Type II Error / Miss)</i>
Actual: False (0)	False Positive (FP)	

The Confusion Matrix - the traditional way

	Predicted: True (1)	Predicted: False (0)
Actual: True (1)	True Positive (TP) <i>(Hit)</i>	False Negative (FN) <i>(Type II Error / Miss)</i>
Actual: False (0)	False Positive (FP) <i>(Type I Error / False Alarm)</i>	True Negative (TN) <i>(Correct Rejection)</i>

For example, if a model predicts if a patient is sick:

- **Rows (Actual):**

The Confusion Matrix - the traditional way

	Predicted: True (1)	Predicted: False (0)
Actual: True (1)	True Positive (TP) (Hit)	False Negative (FN) (Type II Error / Miss)
Actual: False (0)	False Positive (FP) (Type I Error / False Alarm)	True Negative (TN) (Correct Rejection)

For example, if a model predicts if a patient is sick:

- **Rows (Actual):**

- Top Row: All sick patients ($TP + FN$).
- Bottom Row: All healthy patients ($FP + TN$).

- **Columns (Predicted):**

The Confusion Matrix - the traditional way

	Predicted: True (1)	Predicted: False (0)
Actual: True (1)	True Positive (TP) (Hit)	False Negative (FN) (Type II Error / Miss)
Actual: False (0)	False Positive (FP) (Type I Error / False Alarm)	True Negative (TN) (Correct Rejection)

For example, if a model predicts if a patient is sick:

- **Rows (Actual):**

- Top Row: All sick patients ($TP + FN$).
- Bottom Row: All healthy patients ($FP + TN$).

- **Columns (Predicted):**

- Left Column: Everyone the model *said* was sick.
- Right Column: Everyone the model *said* was healthy.

The Confusion Matrix - The metrics melting pot

Sources: [16][17][18][19][20][21][22][23] view · talk · edit

		Predicted condition			
		Predicted positive	Predicted negative	Informedness, bookmaker informedness (BM) $= \text{TPR} + \text{TNR} - 1$	Prevalence threshold (PT) $= \frac{\sqrt{\text{TPR} \times \text{FPR}} - \text{FPR}}{\text{TPR} - \text{FPR}}$
Actual condition	Total population $= P + N$				
	Real Positive (P) ^[a]	True positive (TP), hit ^[b]	False negative (FN), miss, underestimation	True positive rate (TPR), recall, sensitivity (SEN), probability of detection, hit rate, power $= \frac{\text{TP}}{P} = 1 - \text{FNR}$	False negative rate (FNR), miss rate, type II error ^[c] $= \frac{\text{FN}}{P} = 1 - \text{TPR}$
	Real Negative (N) ^[d]	False positive (FP), false alarm, overestimation	True negative (TN), correct rejection ^[e]	False positive rate (FPR), probability of false alarm, fail-out, type I error ^[f] $= \frac{\text{FP}}{N} = 1 - \text{TNR}$	True negative rate (TNR), specificity (SPC), selectivity $= \frac{\text{TN}}{N} = 1 - \text{FPR}$
		Positive predictive value (PPV), precision $= \frac{\text{TP}}{\text{TP} + \text{FP}} = 1 - \text{FDR}$	False omission rate (FOR) $= \frac{\text{FN}}{\text{TN} + \text{FN}} = 1 - \text{NPV}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$
		False discovery rate (FDR) $= \frac{\text{FP}}{\text{TP} + \text{FP}} = 1 - \text{PPV}$	Negative predictive value (NPV) $= \frac{\text{TN}}{\text{TN} + \text{FN}} = 1 - \text{FOR}$	Markedness (MK), deltaP (Δp) $= \text{PPV} + \text{NPV} - 1$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$
		F_1 score $= \frac{2 \text{PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}} = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}$	Fowkes–Mallows index (FM) $= \sqrt{\text{PPV} \times \text{TPR}}$	phi or Matthews correlation coefficient (MCC) $= \frac{\sqrt{\text{TPR} \times \text{TNR} \times \text{PPV} \times \text{NPV}}}{\sqrt{1 - \text{FNR} \times \text{FPR} \times \text{FOR} \times \text{FDR}}}$	Threat score (TS), critical success index (CSI), Jaccard index $= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}$

Figure: From Wikipedia

The Confusion Matrix - The essentials

	Predicted		Row Metrics
	Positive	Negative	
Actual Positive	TP	FN	→ TPR (Sensitivity) $\frac{TP}{TP+FN}$
Actual Negative	FP	TN	→ TNR (Specificity) $\frac{TN}{TN+FP}$
Column Metrics	↓ $\frac{TP}{TP+FP}$ PPV	↓ $\frac{TN}{TN+FN}$ NPV	

- **Row Formulas:** Divide the cell by the **Row Total** (Actual).
- **Column Formulas:** Divide the cell by the **Column Total** (Prediction).

Accuracy

Definition

Accuracy measures the proportion of correct predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Interpretation

It is the probability that the classifier makes a correct decision.

Caution

Accuracy can be misleading for imbalanced datasets, where one class is much more frequent than the other.

Measure what you value

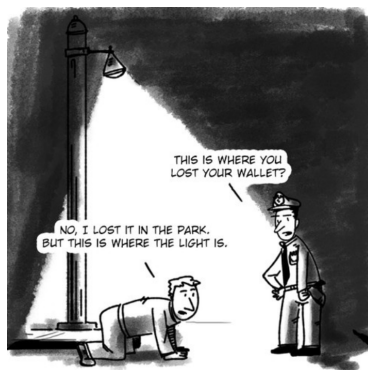


Figure: And if you can't, keep it simple, clear, and correlated

Are These Metrics Clinically Meaningful?

The Problem with Standard Metrics

Metrics like Accuracy and F1 do not let us optimize for what really matters clinically. They fail to answer the critical question: **Is a False Negative worse than a False Positive?**

Are These Metrics Clinically Meaningful?

The Problem with Standard Metrics

Metrics like Accuracy and F1 do not let us optimize for what really matters clinically. They fail to answer the critical question: **Is a False Negative worse than a False Positive?**

But in Medicine...

Scenario A: Cancer Screening

- **False Negative:** Fatal.
- **False Positive:** Biopsy cost + Anxiety.
- *Asymmetry:*

Are These Metrics Clinically Meaningful?

The Problem with Standard Metrics

Metrics like Accuracy and F1 do not let us optimize for what really matters clinically. They fail to answer the critical question: **Is a False Negative worse than a False Positive?**

But in Medicine...

Scenario A: Cancer Screening

- **False Negative:** Fatal.
- **False Positive:** Biopsy cost + Anxiety.
- **Asymmetry:**
 $Cost(FN) \gg Cost(FP)$.

Are These Metrics Clinically Meaningful?

The Problem with Standard Metrics

Metrics like Accuracy and F1 do not let us optimize for what really matters clinically. They fail to answer the critical question: **Is a False Negative worse than a False Positive?**

But in Medicine...

Scenario A: Cancer Screening

- **False Negative:** Fatal.
- **False Positive:** Biopsy cost + Anxiety.
- **Asymmetry:**
 $Cost(FN) \gg Cost(FP)$.

Scenario B: High-Risk Surgery

- **False Negative:** Watchful waiting.
- **False Positive:** Unnecessary chest cracked open.
- **Asymmetry:**

Are These Metrics Clinically Meaningful?

The Problem with Standard Metrics

Metrics like Accuracy and F1 do not let us optimize for what really matters clinically. They fail to answer the critical question: **Is a False Negative worse than a False Positive?**

But in Medicine...

Scenario A: Cancer Screening

- **False Negative:** Fatal.
- **False Positive:** Biopsy cost + Anxiety.
- **Asymmetry:**
 $Cost(FN) \gg Cost(FP)$.

Scenario B: High-Risk Surgery

- **False Negative:** Watchful waiting.
- **False Positive:** Unnecessary chest cracked open.
- **Asymmetry:**
 $Cost(FP) \gg Cost(FN)$.

*We need a metric that accounts for **Consequences**.*

Formalizing Utility: The Cost Matrix

Following the example of a disease prediction, we define a related cost (that could be negative) for each type of prediction (TP, FN, FP or TN). *Utility is the same concept viewed positively — higher utility simply means lower cost.*

	Predict: Disease (1)	Predict: Healthy (0)
Actual: Disease (1)	C_{TP} (Cost of Treatment)	C_{FN} (Cost of Untreated Disease)
Actual: Healthy (0)	C_{FP} (Cost of Side Effects)	C_{TN} (Cost of Reassurance / No Action)

Formalizing Utility: The Cost Matrix

Following the example of a disease prediction, we define a related cost (that could be negative) for each type of prediction (TP, FN, FP or TN). *Utility is the same concept viewed positively — higher utility simply means lower cost.*

	Predict: Disease (1)	Predict: Healthy (0)
Actual: Disease (1)	C_{TP} (Cost of Treatment)	C_{FN} (Cost of Untreated Disease)
Actual: Healthy (0)	C_{FP} (Cost of Side Effects)	C_{TN} (Cost of Reassurance / No Action)

Simplifying the Costs - Assumptions

Often, we set $C_{TP} \approx 0$ and $C_{TN} \approx 0$ and focus on mistakes:

- C_{FN} : How costly is a false negative?
- C_{FP} : How costly is a false positive?

Total Expected Cost

Idea

The **total expected cost** is the average cost we expect to pay under the model's predictions.

From the Confusion Matrix (empirical)

Let $N = TP + TN + FP + FN$. Then

$$\mathbb{E}[\text{Cost}] \approx \frac{1}{N} (C_{TP} TP + C_{TN} TN + C_{FP} FP + C_{FN} FN).$$

Often, $C_{TP} \approx 0$ and $C_{TN} \approx 0$, so the dominant terms are

$$\mathbb{E}[\text{Cost}] \approx \frac{1}{N} (C_{FP} FP + C_{FN} FN).$$

Part 2

Soft Classification

- 1 Hard Classification
 - Definition
 - Confusion Matrix
 - Utility
- 2 Soft Classification
 - Definition
 - Score vs Probabilities
 - Bayes Optimal Classifier
 - Decision Curve Analysis
 - Discrimination vs Calibration
 - AUROC and its limitations

Motivation: From Hard Rules to Soft Scores

So far, we have treated classifiers as "Black Boxes" that output a decision:

$$f(x) \rightarrow \{0, 1\}$$

Why is this insufficient for Healthcare?

Motivation: From Hard Rules to Soft Scores

So far, we have treated classifiers as "Black Boxes" that output a decision:

$$f(x) \rightarrow \{0, 1\}$$

Why is this insufficient for Healthcare?

- **Nuance is lost:** A patient with 51% risk is treated the same as one with 99% risk.
- **Thresholds are fixed:** We cannot adjust for changing resource constraints (e.g., "only 10 ICU beds left").
- **Calibration is unknown:** We don't know if the model is "confident" or just "guessing."

Motivation: From Hard Rules to Soft Scores

So far, we have treated classifiers as "Black Boxes" that output a decision:

$$f(x) \rightarrow \{0, 1\}$$

Why is this insufficient for Healthcare?

- **Nuance is lost:** A patient with 51% risk is treated the same as one with 99% risk.
- **Thresholds are fixed:** We cannot adjust for changing resource constraints (e.g., "only 10 ICU beds left").
- **Calibration is unknown:** We don't know if the model is "confident" or just "guessing."

The "Soft" Approach

Modern ML models (Logistic Regression, Neural Networks, GBMs) natively output a **continuous score** or **probability**:

$$s(x) \in \mathbb{R} \quad \text{or} \quad p(x) \in [0, 1]$$

We convert this score to a hard decision *later*, based on clinical utility.

Scores vs. Probabilities

1. The Raw Score $s(x)$

Most classifiers (SVMs, Neural Nets, Gradient Boosting) internally calculate a raw score:

- **Examples:** Distance to hyperplane, Log-odds (logits).
- **Range:** $(-\infty, +\infty)$.
- **Meaning:** Relative confidence. If $s(A) > s(B)$, patient A is *more likely* to be sick than B.

2. The Mapping to Probability $\hat{p}(x)$ To interpret this as a risk, we squash the score into $[0, 1]$ (e.g., via Sigmoid/Softmax):

$$\hat{p}(y = 1|x) = \sigma(s(x)) = \frac{1}{1 + e^{-s(x)}}$$

The Warning

Just because a number is between 0 and 1 does **not** make it a probability! *A model can say "0.99" for everyone and be wrong 50% of the time.*

Probabilistic Formulation: Defining Risk

1. The Loss Function $L(y, \hat{y})$

- Quantifies the "cost" of a single prediction.
- **0-1 Loss (Standard Accuracy):**

$$L(y, \hat{y}) = \mathbb{I}(y \neq \hat{y}) = \begin{cases} 0 & \text{if } y = \hat{y} \\ 1 & \text{if } y \neq \hat{y} \end{cases}$$

Probabilistic Formulation: Defining Risk

1. The Loss Function $L(y, \hat{y})$

- Quantifies the "cost" of a single prediction.
- 0-1 Loss (Standard Accuracy):**

$$L(y, \hat{y}) = \mathbb{I}(y \neq \hat{y}) = \begin{cases} 0 & \text{if } y = \hat{y} \\ 1 & \text{if } y \neq \hat{y} \end{cases}$$

2. The Risk $R(h)$

- The **Expected Loss** over the joint distribution $P(X, Y)$.

$$R(h) = \mathbb{E}_{X,Y}[L(Y, h(X))]$$

- Interpretation:* The probability that the classifier makes an error on a randomly chosen patient.

Minimizing 0-1 Loss: The Bayes Optimal Classifier

To minimize the risk with the 0-1 loss, the optimal strategy is to select the class with the highest posterior probability for every input x .

The Bayes Optimal Classifier $h^*(x)$

$$h^*(x) = \operatorname{argmax}_{y \in \{0,1\}} P(Y = y | X = x)$$

In the Binary Case ($y \in \{0, 1\}$):

Minimizing 0-1 Loss: The Bayes Optimal Classifier

To minimize the risk with the 0-1 loss, the optimal strategy is to select the class with the highest posterior probability for every input x .

The Bayes Optimal Classifier $h^*(x)$

$$h^*(x) = \operatorname{argmax}_{y \in \{0,1\}} P(Y = y | X = x)$$

In the Binary Case ($y \in \{0, 1\}$):

- Since probabilities sum to 1, "highest probability" implies > 0.5 .

$$h^*(x) = \begin{cases} 1 & \text{if } P(Y = 1 | X = x) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Note: This 0.5 threshold is only optimal because 0-1 Loss treats FP and FN as equal costs.

Bayes Irreducible Error (Bayes Error Rate)

Even with the optimal Bayes decision rule, errors can remain because the class distributions overlap.

Bayes (irreducible) error rate:

$$R^* = \inf_f \mathbb{E}[1\{h^*(X) \neq Y\}] = \mathbb{E}\left[1 - \max_{y \in \{0,1\}} P(Y = y \mid X)\right]$$

- R^* is the *minimum achievable* misclassification probability under the true data distribution.
- If $p(x \mid 0)$ and $p(x \mid 1)$ overlap, then $R^* > 0$: some points are inherently ambiguous.
- Any additional error beyond R^* comes from estimation/approximation (model and finite-data effects).

Bayes Optimal Classifier

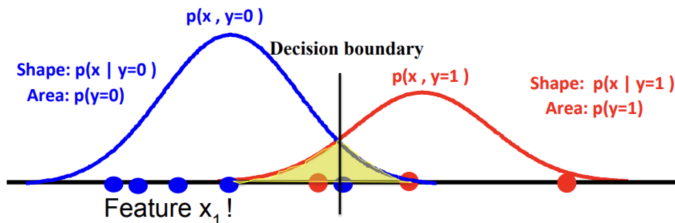


Figure: Visualizing the decision boundary

- Since $p(x)$ is the same for both classes at fixed x ,

Bayes Optimal Classifier

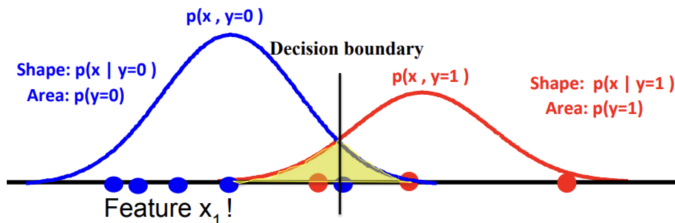


Figure: Visualizing the decision boundary

- Since $p(x)$ is the same for both classes at fixed x ,

$$\arg \max_k P(y = k | x) = \arg \max_k p(x | y = k) P(y = k).$$

- Equivalently, compare joint heights at x :

$$p(x, y = 1) \gtrless p(x, y = 0).$$

Cost-Sensitive Loss Function

If costs are unequal ($C_{FN} \neq C_{FP}$), the standard threshold of 0.5 is wrong. We allow different costs for false positives and false negatives.

$$L(y, \hat{y}) = \begin{cases} 0 & y = \hat{y} \\ C_{FP} & y = 0, \hat{y} = 1 \\ C_{FN} & y = 1, \hat{y} = 0 \end{cases}$$

- $y \in \{0, 1\}$: true label
- $\hat{y} \in \{0, 1\}$: predicted label

Risk and Conditional Risk

Overall (true) risk:

$$R(f) = \mathbb{E}[L(Y, f(X))]$$

Conditional risk of predicting $\hat{y}$ at feature value x :

$$R(\hat{y} | x) = \mathbb{E}[L(Y, \hat{y}) | X = x]$$

To minimize $R(f)$, it suffices to minimize $R(\hat{y} | x)$ for every x .

Conditional Risk for Each Prediction

If we predict $\hat{y} = 1$:

$$R(\hat{y} = 1 | x) = L(0, 1)P(Y = 0 | x) + L(1, 1)P(Y = 1 | x) = C_{\text{FP}} P(Y = 0 | x)$$

If we predict $\hat{y} = 0$:

$$R(\hat{y} = 0 | x) = L(1, 0)P(Y = 1 | x) + L(0, 0)P(Y = 0 | x) = C_{\text{FN}} P(Y = 1 | x)$$

Bayes decision rule:

$$f^*(x) = \arg \min_{\hat{y} \in \{0, 1\}} R(\hat{y} | x)$$

Deriving the Optimal Threshold

Choose $\hat{y} = 1$ whenever it has smaller conditional risk:

$$C_{FP}P(Y = 0 | x) < C_{FN}P(Y = 1 | x)$$

Let $p(x) = P(Y = 1 | x)$. Then $P(Y = 0 | x) = 1 - p(x)$, so:

$$C_{FP}(1 - p(x)) < C_{FN}p(x)$$

$$C_{FP} < (C_{FP} + C_{FN})p(x)$$

$$p(x) > \frac{C_{FP}}{C_{FP} + C_{FN}}$$

Conclusion: Optimal Threshold

Define the optimal threshold

$$\tau = \frac{C_{FP}}{C_{FP} + C_{FN}}$$

Then the Bayes-optimal cost-sensitive classifier is:

$$\hat{y}(x) = \begin{cases} 1 & \text{if } P(Y = 1 | x) > \tau \\ 0 & \text{otherwise.} \end{cases}$$

Conclusion: Optimal Threshold

Define the optimal threshold

$$\tau = \frac{C_{FP}}{C_{FP} + C_{FN}}$$

Then the Bayes-optimal cost-sensitive classifier is:

$$\hat{y}(x) = \begin{cases} 1 & \text{if } P(Y = 1 | x) > \tau \\ 0 & \text{otherwise.} \end{cases}$$

Clinical Intuition

- missing a case is **9x worse** than a false positive ($C_{FN} = 9, C_{FP} = 1$):
 $\tau = \frac{1}{9+1} = 0.10$
- Conclusion:** Treat anyone with $> 10\%$ risk!

From Optimal Threshold to a Decision Rule

Decision problem.

We choose between two *actions/strategies*:

T : take action (e.g. treat / intervene), N : do not act (e.g. no treat / no intervene).

The true state is $Y \in \{0, 1\}$ (e.g. $Y = 1$ diseased, $Y = 0$ not diseased).

Bayes decision rule. Treat when T has lower expected cost than N :

$$\mathbb{E}[C \mid T, x] < \mathbb{E}[C \mid N, x].$$

Equivalently, define the *incremental loss* of choosing T instead of N :

$$\Delta C_1 = C_{TP} - C_{FN}, \quad \Delta C_0 = C_{FP} - C_{TN},$$

and treat when

$$P(Y = 1 \mid x) \Delta C_1 + (1 - P(Y = 1 \mid x)) \Delta C_0 < 0.$$

Bayes-Optimal Threshold and the DCA Special Case

The inequality

$$P(Y = 1 | x) \Delta C_1 + (1 - P(Y = 1 | x)) \Delta C_0 < 0$$

is equivalent to a posterior threshold rule:

$$\hat{y}(x) = 1\{P(Y = 1 | x) > \tau\},$$

$$\tau = \frac{\Delta C_0}{\Delta C_0 - \Delta C_1} = \frac{C_{FP} - C_{TN}}{(C_{FP} - C_{TN}) + (C_{FN} - C_{TP})}.$$

In odds form,

$$\frac{\tau}{1 - \tau} = \frac{\Delta C_0}{-\Delta C_1} = \frac{C_{FP} - C_{TN}}{C_{FN} - C_{TP}}.$$

Special case: $C_{TN} = 0$ and $C_{TP} = 0$. Then $\Delta C_0 = C_{FP}$ and $-\Delta C_1 = C_{FN}$, so

$$\tau = \frac{C_{FP}}{C_{FP} + C_{FN}} \quad \text{and} \quad \frac{\tau}{1 - \tau} = \frac{C_{FP}}{C_{FN}}.$$

Thus the DCA weight matches the Bayes-optimal cost ratio under this common assumption.

Outcomes at a Given Threshold τ

For a dataset of size N , applying threshold τ gives:

$$TP(\tau), \quad FP(\tau), \quad TN(\tau), \quad FN(\tau).$$

Let

P = number of true positives in the population,

$\pi = \frac{P}{N}$, the prevalence (proportion of positives).

We now want to evaluate how good a decision rule is at each τ .

Intuition: What is Net Benefit? (relative to a baseline)

Decision Curve Analysis asks:

Compared to a simple baseline policy (e.g. treat-none), does using the model reduce expected loss/increase expected utility?

Take **treat-none** as the baseline. Relative to this baseline:

- Each treated true case (TP) *avoids* a missed-treatment loss of size C_{FN} .
- Each treated non-case (FP) *incurs* an unnecessary-treatment loss of size C_{FP} .

The incremental value per person (up to a positive scaling) is:

$$\frac{TP}{N} - \frac{FP}{N} \cdot \frac{C_{FP}}{C_{FN}} = \frac{TP}{N} - \frac{FP}{N} \cdot \frac{\tau}{1 - \tau}.$$

Intuition: What is Net Benefit? (relative to a baseline)

Decision Curve Analysis asks:

Compared to a simple baseline policy (e.g. treat-none), does using the model reduce expected loss/increase expected utility?

Take **treat-none** as the baseline. Relative to this baseline:

- Each treated true case (TP) *avoids* a missed-treatment loss of size C_{FN} .
- Each treated non-case (FP) *incurs* an unnecessary-treatment loss of size C_{FP} .

The incremental value per person (up to a positive scaling) is:

$$\frac{TP}{N} - \frac{FP}{N} \cdot \frac{C_{FP}}{C_{FN}} = \frac{TP}{N} - \frac{FP}{N} \cdot \frac{\tau}{1 - \tau}.$$

Net Benefit. DCA defines the bracketed quantity as *net benefit*:

$$NB(\tau) = \frac{TP}{N} - \frac{FP}{N} \cdot \frac{\tau}{1 - \tau},$$

Reference Strategies in Decision Curve Analysis

We compare the model to two simple strategies.

Treat none (predict all 0):

$$NB_{\text{none}}(\tau) = 0$$

Treat all (predict all 1):

$$NB_{\text{all}}(\tau) = \pi - (1 - \pi) \frac{\tau}{1 - \tau}, \quad \pi = \frac{P}{N}.$$

- π = prevalence of the outcome
- A model is useful when $NB_{\text{model}}(\tau)$ exceeds both reference curves.

The Decision Curve

The decision curve plots

$$\tau \longmapsto NB_{\text{model}}(\tau)$$

and compares it to $NB_{\text{none}}(\tau)$ and $NB_{\text{all}}(\tau)$.

At a given threshold τ :

- If $NB_{\text{model}}(\tau)$ is highest $\rightarrow$ using the model gives the most net benefit.
- If NB_{none} is highest $\rightarrow$ better to treat nobody.
- If NB_{all} is highest $\rightarrow$ better to treat everybody.

This shows for which risk thresholds a model is clinically or practically useful.

Decision Curves - Example

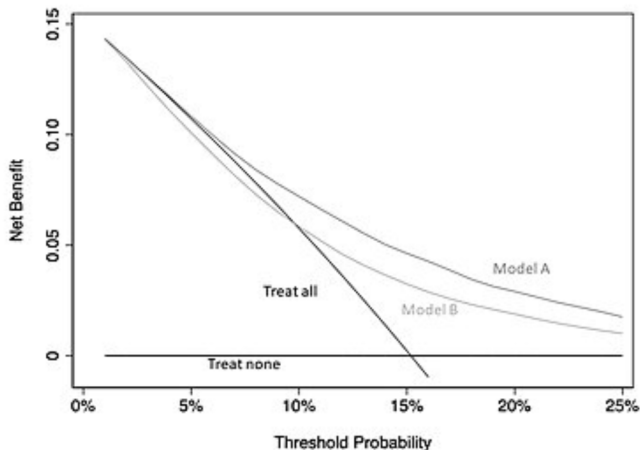


Figure: Comparing the three strategies. For low threshold probabilities, the models are not better than a treat all approach.

From Decision Utility to Model Performance

Decision Curve Analysis evaluates **clinical/practical usefulness**:

$$\text{Net Benefit}(\tau)$$

which depends on a chosen threshold τ and a harm–benefit tradeoff.

We now shift focus from *decisions* to *predictions*:

How good is the model itself, independent of a specific decision threshold?

This leads to two key properties:

- **Discrimination** — can the model separate positives from negatives?
- **Calibration** — are predicted probabilities numerically correct?

Setup and Notation for the coming slides

- We consider a binary classification dataset (X, y) with labels

$$y_i \in \{0, 1\}.$$

- A model $f_\theta : \mathcal{X} \rightarrow \mathbb{R}$ maps an input x to a real-valued score:

$$x \mapsto s = f_\theta(x), \quad s \in \mathbb{R}.$$

- Define random variables X, Y, S for inputs, labels, and scores:

$$S = f_\theta(X),$$

with probability density (or mass) functions p_X, p_Y, p_S .

- Define class-conditional score random variables:

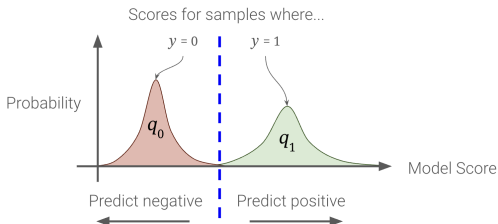
$$S_1 := S \mid (Y = 1), \quad S_0 := S \mid (Y = 0),$$

with densities

$$q_1(s) = p_{S|Y}(s \mid 1), \quad q_0(s) = p_{S|Y}(s \mid 0).$$

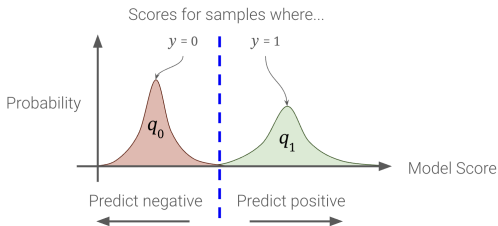
Discrimination vs Calibration

- Discrimination: Can my model tell positive classes apart from negative classes?

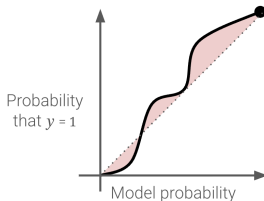


Discrimination vs Calibration

- Discrimination: Can my model tell positive classes apart from negative classes?

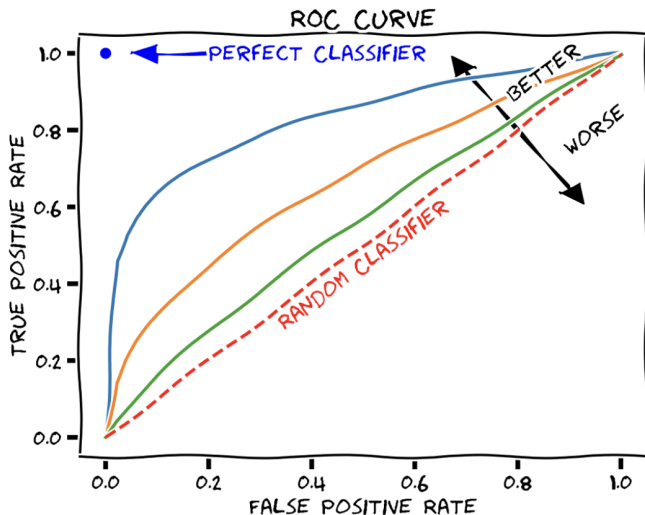


- Calibration: Are the probabilities that my model outputs reliable? How close is S to $p_{Y|X}$?



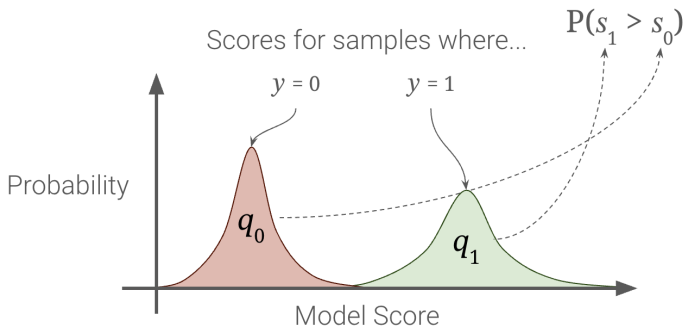
AUROC - How not to define it

AUROC = Area Under the Receiver-Operator Curve



AUROC - a better perspective

AUROC = the probability that a randomly chosen positive sample will receive a higher score than a randomly chosen negative sample



AUROC is a **threshold-independent** measure of discrimination (ranking ability).

AUC Limitations: Simpson's Paradox

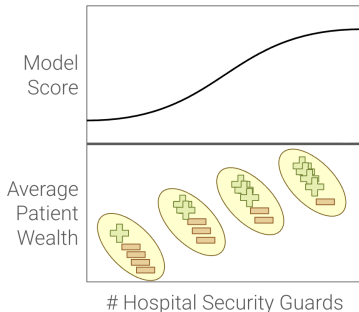
Key point. A model can have a strong overall AUROC while having **AUROC** < 0.5 within every subpopulation.

AUC Limitations: Simpson's Paradox

Key point. A model can have a strong overall AUROC while having **AUROC** < 0.5 within every subpopulation.

Mechanism (Simpson's paradox).

- **Confounder:** average patient wealth.
- Wealthier hospitals tend to have $\uparrow$ guards and $\uparrow$ health.
- Aggregating across wealth groups makes “more guards $\Rightarrow$ healthier” look true, so overall AUROC looks good.
- **Within each wealth group**, the association reverses (guards track severity), so ranking can be worse than random.



Overall trend differs from within-group trends (Simpson's paradox).

Evaluating Models Under the Intended Conditions of Use

To avoid Simpson-type failures, evaluate the model in the **same conditions in which it will be used**.

- If you don't have a deployment scenario planned, and this is for method development or benchmarking, etc., then use AUC and/or calibration measures. Nothing else is that relevant.
- If you have a deployment target in mind, explicitly simulate the utility function(s) you would have in that setting. Nothing else is relevant.

Discrimination & Calibration: Brier Score

The **Brier score** measures how close predicted probabilities are to the true outcomes.

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2$$

- $\hat{y}_i$: predicted probability $P(Y = 1 \mid x_i)$
- $y_i \in \{0, 1\}$: true outcome

What it measures:

- Penalizes inaccurate probability estimates
- Reflects both **calibration** (are probabilities correct?) and **discrimination** (are positives assigned higher risk?)

Lower Brier score indicates better probabilistic predictions.

Calibration: Expected Calibration Error (ECE)

ECE measures how well predicted probabilities match observed outcome frequencies.

Idea:

- Group predictions into bins by predicted risk
- Compare average predicted probability to observed event rate in each bin

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\text{acc}(b) - \text{conf}(b)|$$

- $\text{acc}(b)$: fraction of positives in bin b
- $\text{conf}(b)$: average predicted probability in bin b

Key point:

- ECE measures **calibration only**
- It does *not* measure ranking or discrimination

Lower ECE indicates better calibrated probability estimates.

Summary - How These Pieces Fit Together

- **Discrimination** → can the model separate high-risk from low-risk individuals?
Example: AUROC
- **Calibration** → are predicted risks trustworthy?
Example: ECE
- **Decision Curve Analysis** → visualize the possible utility under different assumed harm benefit ratios.
Example: Net Benefit

Good discrimination + good calibration $\implies$ more reliable risk thresholds $\implies$ higher potential net benefit in DCA

Additional resources:

- General classification metrics
https://scikit-learn.org/stable/modules/model_evaluation.html
- AUROC limitations <https://neurips.cc/virtual/2024/poster/95133>
- Decision Curve Analysis
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2577036/>

Thanks! Any questions?